River Deep Mountain AI

D2.3 – Insights report from stakeholder engagement

WP2: Stakeholder Engagement and Outreach
Phase 2 & 3

River Deep Mountain AI

Version overview

Version	Date	Authors	Comments
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Sign-off

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1 Overview

This document outlines the key findings from Phase 2 (model development) and 3 (model validation and benchmarking) of the River Deep Mountain AI (RDMAI) project, specifically related to Work Programme 2 (WP2 - Stakeholder Engagement and Outreach).

Phase 1 summary

As detailed in the Phase 1 '*Insights report from stakeholder research*', WP2 initially focussed on defining the key themes and building blocks that the remainder of the project would focus on. Following a workshop at the 2024 Northumbrian Innovation Festival, 66 stakeholders from 25 different organisations were invited to semi-structured interviews. Through these interviews, we aimed to ensure that RDMAI model development focused on the themes that would best address stakeholder needs. The eight overarching themes identified through these stakeholder interviews were refined to four key themes, via a prioritisation framework that focussed on 'potential impact' and 'model feasibility'. The themes provided the starting point for Phase 2 of the wider project, from which model development and testing could commence.

Summary of phases 2 and 3

The stakeholder engagement during Phases 2 and 3 primarily focussed on maintaining stakeholder input and feedback cycles during the model development, testing, and validation phases of work. This helped to ensure that stakeholders could keep up to date with model progress, influence key model development decisions, and feedback any concerns about model development, ensuring the final model outputs remained relevant and useful to interested parties.

This report summarises how stakeholders were engaged during Phases 2 and 3 of the project and describes how the two main deliverables of WP2 were met:

1. Wider stakeholder share, occurring quarterly (T2.5)**:** - *Session with Stream, CastCo, Mainstreaming NbS and wider stakeholders (catchment management) to guide and validate development of machine learning models (October 2024–January 2026).*

2. Validate the value of machine learning objectives with stakeholders (T2.6)**:** *Continuously conduct validation sessions and additional research with stakeholders to validate and test the value of our machine learning models. Happens in phases 2 and phase 3. (January 2025–October 2025).*

Terms

The terms referred to in this report are outlined below, and the relationship between the terms is illustrated by Figure 1. The definitions of these terms have not changed since phase 1 of the project.

Theme – *An overarching topic that has been identified as being of interest to stakeholders.*

Building block – *The defined themes are broad in nature; to address them effectively, they are broken down into more specific, subcategories termed 'AI/ML building blocks'. The building blocks within each of the themes detail the ways in which each theme could be 'modelled'. The AI/ML building blocks represent tangible goals aligned with each theme and serve to focus efforts on distinct, solvable challenges. The AI/ML building blocks are essentially modelling approaches, that help highlight the ways in which a theme can be addressed.*

Use case – *Outlines how a stakeholder could use model outputs. A use case might be made up of more than one building block.*

Artificial Intelligence (AI) – *Artificial intelligence is a field of science concerned with building computers and machines that can reason, learn, and act in such a way that would normally require human intelligence or that involves data whose scale exceeds what humans can analyze[1]. Machine Learning (ML) is a focused branch of AI, where systems improve their accuracy through data exposure rather than explicit programming for each scenario. Trained on vast datasets, ML models detect patterns, predict outcomes, and classify information.*

Modelling – *Throughout this report, we refer to AI/ML, models and modelling. Unless otherwise specified, all these refer to the training and development of artificial intelligence using different techniques from the field of AI, including machine learning, such as deep learning, neural networks and more.*

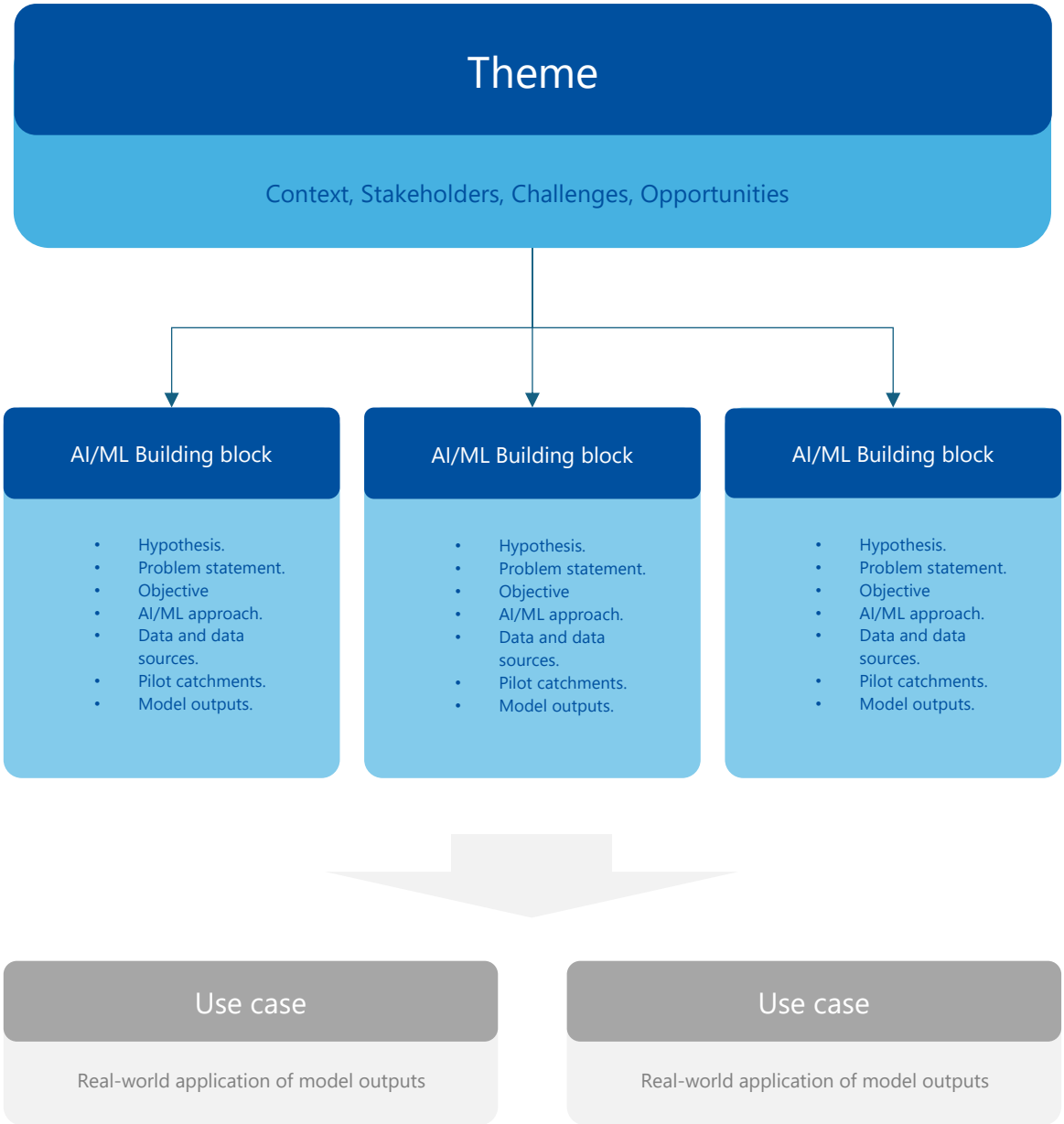

Figure 1: Taxonomy of themes, building blocks and use cases.

[1] https://cloud.google.com/learn/what-is-artificial-intelligence

2 Stakeholder engagement

Throughout phases 2 and 3, stakeholders were engaged through a variety of different mechanisms. An overview of these different engagement mechanisms is provided below.

Theme leads

To help facilitate the interaction between data scientists and domain experts (both from internal project partners and wider industry stakeholders), 'theme leads' were assigned to each of the three core themes:

- Theme 1: Modelling high-priority pollutants – Nicolai Tarp (Cognizant)
- Theme 2: Optimising catchment monitoring – Dan Pearce (WRc)
- Theme 3: Getting more value from continuous monitoring – Elisabeth Sund (Cognizant) / Gustavo Ostos Rios (Cognizant)

Theme leads were responsible for coordinating 'bi-weekly domain calls', ensuring relevant domain experts (both internal project partners and external stakeholders) had the opportunity to contribute to key model development decisions wherever possible and ensuring feedback from stakeholders was incorporated into the model development process where appropriate.

Bi-weekly domain calls

During phase 1, large, monthly Technical Advisory Group (TAG) sessions were used to gather input from wider domain expertise and help guide key technical decisions. However, early on in phase 2, it became evident that these pre-defined governance structures and high-level advisory groups were inefficient and did not always encourage active, open dialogue across all project partners. Stakeholder engagement within these larger TAG sessions was limited and it did not support the fast-paced, agile decision-making process needed within the RDMAI project approach.

Consequently, the decision was made to move away from large TAG sessions and instead focus on smaller, shorter, bi-weekly domain calls for each of the different building blocks. These smaller, more targeted workshop-style sessions allowed the project team to get the most value out of the limited domain expertise time available. All TAG members were invited to participate in whichever building block sessions they wished to attend. Over the course of phase 2 and 3, the people attending each of the bi-weekly meetings varied depending on the engagement of stakeholders and the needs of the model development team. Inactive TAG members or stakeholders were not included in all correspondence. Wider TAG sessions would

still take place to provide a high-level overview of progress across the project and acted as a form of stakeholder engagement to help ensure each of the models met stakeholder needs.

Although the smaller bi-weekly domain calls worked better than the wider TAG sessions, the level of engagement from stakeholders in these bi-weekly calls did vary. In general, non-costed project partners / stakeholders often struggled to regularly attend these calls, due to limited availability and other commitments. Similarly, relevant documentation and model outputs were shared with the wider TAG group at several points across the project, to try and gather input into model development However, much like the bi-weekly calls, feedback received was minimal from non-costed partners. Consequently, future projects should consider making relevant stakeholders costed partners (even if the costs are low), in order to encourage a high level of technical engagement and accountability.

Demonstration days

To ensure external stakeholders (that were not actively involved within the bi-weekly domain calls) remained informed on model development and had an opportunity to feedback on model progress and direction, quarterly stakeholder sessions (i.e., '*demo days*') were organised. The *demo days* followed a conference-style format, allowing attendees to join virtual sessions based on their specific interests in particular themes or building blocks. These sessions were open to all, enabling anyone to participate by registering in advance. The first two *demo days* were advertised through our existing project network from the TAG and other partners who had shown prior interest. The third *demo day* and the final project showcase were organised by Spring and advertised across they're pre-established networks. The session structure included a presentation about model development for each building block, followed by a corresponding Q&A session. Over the course of phase 2 and 3, three *demo days* were organised:

1. 16th December 2024 (44 attendees)
2. 6th March 2025 (61 attendees)
3. 30th April 2025 (69 attendees)

The first two *demo days* were coordinated and facilitated by Cognizant, whilst the third was organised by Spring. Following the completion of each demo day, comments and feedback were collated for each of the individual building blocks and incorporated into future model development work. Any outstanding questions/queries/comments that were not addressed during the Q&A sessions were captured by Spring and responded to directly by the relevant theme lead or data scientist. For each of the demo days, a breakdown of attendance by sector can be found in Appendix A.

Showcase Event

On the 25th November 2025, 139 people attended the River Deep Mountain AI Spring Showcase event. During this event, the key outputs, final models, use cases and lessons learnt were presented to attendees. Following the session, 18 individuals requested 'deep dive' sessions to further explore how these models might be applied or adapted to their work. A breakdown of attendance by sector can be found in Appendix A.

In-person collaborative sessions

Several in-person, collaborative workshop-style sessions were undertaken over the course of phases 2 and 3. These sessions provided an opportunity for project partners and other stakeholders engaged with the project to discuss key technical aspects of the model development process and highlight any outstanding issues that needed to be addressed. All project partners were invited to attend the sessions. During these sessions, 'lessons learnt' and 'ways of working' were also discussed to improve the efficiency, transparency and collaboration between the wider project team. A brief overview of each of these in-person collaborative sessions is provided below.

Cambridge: 11–12th February 2025

The first in-person collaboration day was hosted by Xylem in Cambridge, with representatives of the following organisations attending: Northumbrian Water Limited (NWL), Cognizant, Xylem, WRc, ADAS, Spring and The Rivers Trust. The two-day event primarily consisted of full-team 'workshop-style' sessions, reviewing the technical progress that had been made on each of the building block models. Where possible, technical or coding issues were resolved, and key technical decisions were agreed upon to guide further model development. Additionally, any outstanding concerns were flagged by the wider team and fed back to the relevant data science lead for each of the models. Alongside these technical review sessions, some time was spent discussing wider-team 'lessons learnt', to improve cross-organisational communication and encourage more efficient collaboration for the remainder of the project. A more detailed overview of overall project lessons learnt is provided in the final Ofwat report.

Swindon: 7–8th May 2025

The second in-person collaboration day was hosted by WRc in Swindon, with representatives of the following organisations attending: NWL, Cognizant, Xylem, WRc, ADAS, Spring, The Rivers Trust, Dŵr Cymru Welsh Water (DCWW), South West Water and Wessex Water. The first day primarily focussed on 'issue resolution' workshops for each of the different building blocks. During these sessions, project partners were able to raise concerns about the model development. These issues were either addressed on the day or further discussions were scheduled for after the event. The second day mainly focussed on developing the validation plans for each of the building blocks and assigning corresponding roles/responsibilities for these tasks. Towards the end of the second day, stakeholder engagement/implementation and model documentation plans were discussed. The team also attended a site visit to Warleigh Weir and a meeting at Wessex Water offices to discuss the E. *coli* model that Wessex Water had been working.

London: 7–8th October 2025

The third in-person collaboration day was hosted by Cognizant in London, with representatives of the following organisations attending: NWL, Cognizant, Xylem, WRc, ADAS, Spring, The Rivers Trust, South West Water and Wessex Water. The primary objective of the workshop was to review the model development and validation that had taken place during phase 3, as well as discuss model use cases and model adoption. Any outstanding actions or concerns could then be addressed in the following weeks to allow for internal 'model signoffs' to be completed in November. An overview of the model benchmarking plans was also presented and discussed during the second day, along with a final workshop of project retrospectives and lessons learnt. The model benchmarking reports can be found in the project GitHub (https://github.com/Cognizant-RDMAI).

Swindon: 5th February 2026

The fourth in-person collaboration day a hybrid (in-person and virtual) took place on the 5th February, in the WRc office in Swindon. The session scoped out opportunities for further model development and adoption, by taking on board any industry recommendations from stakeholders that have provided feedback about the final RDMAI outputs. Outputs from this session are included in the final project report.

Northumbrian Innovation Festivals

July 2024:

RDMAI launched ahead of its formal start date at the NWL Innovation Festival in July 2024, with a 4-day open invitation innovation sprint attended by over 70 delegates from across the water sector and other technology areas. This explored the original challenge areas, potential use-cases, data availability, and data science methodologies.

July 2025:
In the second NWL innovation festival, we focused on adoption, defining key elements that can help ensure that AI/ML models could be adopted into business-as-usual by relevant stakeholders: water companies, NGOs, community groups or regulators. The sprint was attended by 27 delegates from 19 different organisations across the water sector and other technology areas.

Additional Industry Events

Over the course of the project, the team attended and presented RDMAI outputs and learning at a range of different catchment partnerships and industry events. A number of key participations are included below:

- CIWEM UDG: Daniel Pearce received the Best Presentation Award at the 2025 CIWEM Urban Drainage Group conference.
- EPIC2025: Elisabeth Sund and Nicolai Tarp's paper "Leveraging Collective Intelligence to Improve River and Coastal Health Insights from England and Wales", was published and presented physically in Helsinki.
- RSD14: Elisabeth Sund and Nicolai Tarp's paper "River Deep Mountain AI: A Systemic Design Approach to Cultivating Shared Ownership of River Health in the United Kingdom", was published and presented virtually.
- Sensors in Water Interest Group (SWIG) 2025: Collaborative presentation by Chris Dawson (Xylem) and Kuntal Ganguly (Cognizant)
- WWEM 2025: Chris Dawson presented on RDMAI at leading UK water sector tradeshow
- WEFTEC 2026: Chris Dawson's paper: "River Deep Mountain AI: Open-source AI/ML models for waterbody health" Submitted for consideration pending response.

3 Final iteration of themes and building blocks

The primary outcome from the stakeholder engagement in phase 2 and 3 of RDMAI was the 'reshaping' and continual refinement of the project themes and building blocks. Consequently, and intentionally, the final project themes and building blocks are not identical to those set out in the phase 1 stakeholder insights report, particularly for theme 2 and 3. Additionally, the phase 1 stakeholder insights report set out several suggested pilot catchments that each of the different models would be tested within. However, as a wider project team, it was decided that the model

development should focus on building models that were applicable at the regional or national scale and then these could be tested/applied on a selection of individual catchments during the validation and benchmarking exercises. This helped ensure that the models remained applicable for multiple stakeholders and catchments, rather than producing a set of catchment-specific models that could only be applied in selected circumstances. An updated, final iteration of the project themes and building blocks are included below. Where possible, key examples of how stakeholders influenced model development decisions have been included.

3.1 Theme 1: Modelling high-priority pollutants

The objective of theme 1, modelling high-priority pollutants, did not change considerably during phase 2 and 3. Instead, our continuous engagement with stakeholders confirmed and strengthened our confidence in this theme. The main objective of theme 1 has been to get more out of the data that are already being collected throughout the water sector to fill in gaps in existing data, reduce the need for additional monitoring and reduce the time and cost of detecting high-priority pollutants, such as orthophosphate and *E. coli*.

A summary of the three themes and related building blocks can be found in Table 1. For further details on the background and reasoning behind the theme, please see the D2.3 Phase 1 report.

Theme	**Building Block**
Theme 1 – Modelling high priority pollutants	1A – Use AI/ML to estimate river flow in ungauged water bodies.
	1B – Use AI/ML to predict E. coli to reduce reliance on time-consuming laboratory testing.
	1C - Use AI/ML to estimate orthophosphate to reduce reliance on expensive equipment.
Theme 2 – Optimising catchment monitoring	2A - Use AI/ML and remote sensing to detect potential sources of pollution.
	2C - To develop a high-level water quality risk mapping framework to help inform catchment monitoring and intervention plans.
Theme 3 – Getting more value out of continuous monitoring	3A - Detecting and clustering anomalies in continuous water quality monitoring data.

Table 1: Summary of themes and building blocks.

The objective of theme 1 was originally broken down into four AI/ML building blocks (see D2.3 Phase 1), which we have continued to develop and shape during phase 2 and 3.

The most considerable change has been the decision to not pursue building block 1D (Use AI/ML to estimate the sources of high-priority pollutants). This decision was made early during phase 2, as we were made aware that others had already worked on this outside of our project (namely the Rivers Trust), and we decided to focus our efforts on the first three building blocks instead, knowing that all three building blocks would benefit the work already completed on pollution travel-time outside of RDMAI. In addition, the Open Risk Mapping model developed as part of theme 2 can be used to help indicate sources of high-priority pollutants, meaning much of this originally planned building block was addressed anyway.

On an individual level, there have been some minor changes to some of the building blocks names, objectives, and approach, which will be detailed below.

Building Block 1A: Use AI/ML to estimate river flow in ungauged waterbodies.

The naming of this building block has been adjusted to better convey the focus of our work; estimating river flow in ungauged waterbodies. Theme 1 is entitled 'Modelling high-priority pollutants'. While river flow is not specifically a pollutant, water quality and quantity (river flow) are intrinsically linked. Therefore, prediction of river pollution requires either observed or estimated river flow.

Defined problem and value:
The defined problem and value of building block 1A has remained unchanged:

- Flow data are limited in most catchments. Besides the usefulness of flow predictions for managing water quantity, flow regulates catchment pollutant pathways to surface water, instream dilution of pollutants, and pollutant residence times in surface water. Consequently, flow is a key parameter required to predict water quality, such as in the SAGIS source apportionment model, and is essential to understand where pollution is coming from and pollutant loads.
- Currently, flow data gaps are filled using simulated flow data or through surrogate monitored flow from similar (proximate, of similar size, and/or having similar hydrological characteristics) catchments. In particular, one of the key drivers to improve flow estimations in ungauged catchments was for use in the SAGIS/SIMCAT models widely used by the industry to understand pollutant source apportionment.
- Stakeholders undertake expensive monitoring programmes to gather flow data. Monitoring campaigns like this can easily cost upwards of £300,000 annually (information shared by stakeholder in early interviews).

- Towards the start of model development, there was significant debate about what flow metrics would be most useful for the model to estimate (e.g. average daily flow, Q95, etc.). Following input from project-partner and wider stakeholder technical experts, it was decided to focus on estimating average daily flow, as many other metrics can be calculated from this information (e.g. Q95).

Technical specifications:

- **Modelling techniques**: Based on stakeholder inputs and literature reviews, we decided to use a long short-term memory (LSTM) model, that works particularly well for timeseries data. We used this together with an encoder for static attributes. Using data available from the CAMELS-GB dataset, we included key influencers of flow, such as weather, soil types, land cover, topography, human influences and more (see model output report for more detail). CAMELS-GB was chosen as our primary dataset, as it provided us with a comprehensive dataset early-on, allowing us to move directly into experimenting with AI/ML for estimating flow and testing our hypothesis. Using CAMELS-GB carries the inherent limitation of only being available for catchments with existing flow gauges. However, as the alternative was to undertake a time-consuming task of collecting all intended input data independently, we decided to use CAMELS-GB to give ourselves as much time as possible to experiment and build our proof of concept. To ensure that the model was developed on trustworthy data, we decided to discard the flow data from CAMELS-GB and instead use quality-controlled flow data from the Hydrology Data Explorer in combination with the static and dynamic data provided by CAMELS-GB. While this approach limited our focus to England (instead of Great Britain), we deemed this necessary to ensure trust in the model input. During development, some catchments were found to perform worse than most catchments. Based on stakeholder inputs we concluded that this might be because they are located on top of a bedrock of chalk feeding groundwater back into the river (also known as chalk streams), which are notoriously difficult to model when it comes to flow. Therefore, we decided to include data about known chalk streams. However, this did not seem to improve the model performance. Please see the Model Output Report and Benchmarking report for further details on performance.

- **Data sources**: CAMELS-GB, Hydrology Data Explorer, Natural England (Chalk streams).

Pilot Catchments:

- Phase 2: Trained on all CAMELS-GB catchments in England with available gauged flow in the Hydrology Data Explorer. For phase 3, we decided to validate the model using 'pseudo-ungauged' catchments generated using the

original CAMELS-GB dataset (see Model Output Report for further details on validation approach). In other words, the model was validated on 'unseen' CAMELS-GB catchments, meaning from the perspective of the model, no gauging data was available for these locations.

Outputs:
The model outputs historical daily mean flow (in m^3/sec). This metric can then be applied to infer key flow metrics for a catchment, such as mean flow and flow exceedance percentiles, like Q95 (flow equalled or exceeded 95% of the time).

Building Block 1B: Use AI/ML to predict *E. coli* to reduce reliance on time-consuming laboratory testing.

During phase 2 and 3, we have continuously engaged with subject-matter experts and stakeholders involved in River Deep Mountain AI. For building block 1B specifically, we have also engaged with the team from the Environment Agency working on Pollution Risk Forecasts (PRF)[1]. This engagement provided us with valuable insights into what is already being done from a regulatory standpoint and helped us shape and share our approach.

Defined problem and value:
The defined problem and value of building block 1B has remained unchanged:

- Monitoring *E. coli* is currently constrained by the time it takes to complete a laboratory test and identify *E. coli.* colony forming units per 100ml (CFU/100ml). This is a process that, depending on logistics, takes at least 48 hours and up to a full week. Laboratory measures of *E. coli* are also costly, ranging between £20 and £100 pounds for each sample processed. This cost limits the *E. coli* samples that can be processed across time and space. The time taken to processes *E. coli* samples makes monitoring sub-optimal for use in a real-time or near real-time warning systems.
- *E. Coli* is a high-priority pollutant for regulators to monitor. It plays a key role in classifying and regulating designated bathing waters.
- In most cases, knowing that bathing water contained *E. coli.* 48 hours ago, is not very useful to act or warn.

Technical specifications:

[1] https://environment.data.gov.uk/bwq/profiles/data-prf.html

- **Modelling techniques:** We decided to use the XGBoost machine learning library for developing our model(s) as it showed the best performance across various experiments.
- The RDMAI *E. coli.* model comes in two variations, according to data requirements and output, respectively, resulting in four model variations.
- By data requirements:
    - The **light model** requires CORINE land use data, weather data and remote sensing MODIS data. It has scope for forecasting given that the input parameters can be forecast. It also has greater spatial and temporal scope for application than the advanced model as the input data are available for any point on a river or coast and at a fine temporal resolution.
    - The **advanced model** requires all the data used in the light model as well as certain water quality parameters, including suspended solids, phosphate-P and nitrate-N and nitrite-N. These additional data were obtained from the Environment Agency's water quality monitoring archive. The spatial and temporal constraints of application of this model variant are dependent on the additional water quality parameters required. Also, this model is not suitable for forecasting given its dependence on water quality monitoring data.
- By model output:
    - The **estimator model** provides simulations of E. coli concentrations in Colony Forming Units (CFU)/100 ml.
    - The **classifier model** provides binary outputs: 'lower risk or 'higher risk', based on thresholds of E. coli concentration associated with risk of illness in bathing waters

- **Data inputs:**
- Laboratory samples of *E.* coli levels from the Environment Agency (EA), Natural Resources Wales (NRW) and Scotland Environment Protection Agency (SEPA). Based on conversations with subject-matter experts, it was concluded that the *E. coli* samples from all three regulatory bodies are comparable, and therefore it was deemed that they were all suitable for our model development. EA data were used for all models, whereas NRW and SEPA were only used for the light versions. This was decided as NRW and SEPA data did not have the necessary water quality parameters for the advanced model corresponding with *E. coli* samples.
- Weather data: Historical weather data were acquired using the Open Meteo API (see model output report for details). The inclusion of weather data was debated by internal project and external technical experts. Whilst there were initial concerns that the inclusion of certain weather parameters (such as windspeed) were irrelevant, it was decided that all available weather

parameters should be included, as it was not possible to complete rule out correlation between these parameters and E. *coli* levels.

- Water quality: Water quality parameters such as nitrite, nitrate, water temperature, suspended solids and phosphate were used from the EA Water Quality Archive. These were chosen based on extensive data exploration together with subject-matter experts and availability together with *E. coli* samples.
- Different land cover categories (from CORINE land cover).
- Leaf Area Index (LAI), Fraction of Photosynthetically Active Radiation (FPAR), Land Surface Temperature and Emissivity (from MODIS).
- We did not have time to test the usefulness of tidal, salinity or CSO data, but they were all mentioned by subject-matter experts as potential useful data inputs.

- **Data sources:** EA Water quality archive, NRW Bathing Water samples, SEPA bathing water samples, CORINE land cover, MODIS satellite data, Open Meteo Historical Weather data.

Pilot catchments:

- Phase 2: We decided to train our advanced models on *E. coli* samples from England (EA) sampled together with relevant water quality parameters. In addition, we decided to train a light version of our models on *E. coli* samples from England, Wales and Scotland to increase *E. coli* sample datapoints and spatiotemporal coverage. Phase 3: We decided to validate our models against *E. coli* samples collected in the past year (not included in our original training dataset).
- We decided not to rely on any local *E. coli* datasets, e.g., from Warleigh Weir, as we found it to be more valuable to use national datasets. This decision was made to create a more robust and scalable model, more adaptable to different bathing waters. Models build using only on a single local dataset would be more difficult to scale and require more additional training to be useful at scale. Local datasets have instead been used in our benchmarking exercise (see benchmark report).

Outputs:

Model outputs have been defined more clearly in phase 2. Below is an overview of the outputs from building block 1B:

Output formats:

- **The classification model:** Outputs a binary result estimating the *E. coli* to be either above or below a defined threshold. Based on inputs from subject-matter experts and EA guidelines, we have used 500 CFU/100ml and 1000

CFU/100ml as our thresholds. ≤500 CFU is the threshold for good condition in coastal bathing waters and ≤1000 CFU is the threshold for good condition in inland bathing waters.

- **The estimation model:** Outputs are an estimation of the *E. coli* concentration in CFU/100ml.

"The mix of project partners quickly overcame initial scepticism of the ambitious scope of the project … The project embodied a culture of co-learning and was keen to utilise my technical understanding of the complexity of environmental science as well as disseminate the capabilities of cutting-edge technologies. RDMAI has been truly democratic and has generated open-source tools available for all. The work of RDMAI puts the power to understand our environment into more hands and give us the tools to communicate environmental complexity more accessibly. I am personally excited to utilise the E. coli prediction models within my own work. I firmly believe that this project has been an excellent example of how the water industry wants and needs to work with other sectors to enhance our natural environment."

John Bagnall, Technical Delivery Manager, Wessex Water

Building Block 1C: Use AI/ML to estimate orthophosphate to reduce reliance on expensive equipment.

The main change made to building block 1C, has been the shift in focus from phosphate to orthophosphate. This decision was made during our stakeholder engagement to increase the chance of success for our model. Data availability turned out to be a major issue for modelling phosphate, as phosphate samples only made up 0.2% of the 24 years of data intended for developing this model. To address this, we decided to shift our focus to orthophosphate, a water quality parameter that is also important for understanding waterbody health and like phosphate, a parameter that is expensive to measure. However, orthophosphate samples made up 2% of the 24 years of data, greatly improving the data availability for our model development.

Defined problem and value:
Despite our shift in focus, we ensured that our defined problem and value remained the same:

- Monitoring orthophosphate requires laboratory testing and wet chemistry, and continuous monitoring is only available using very expensive sondes.
- Orthophosphate is a high-priority pollutant for regulators and water companies to monitor and plays a key role in the ecological classification of a waterbody.

Technical specifications:

- **Modelling techniques**: We decided to use a hybrid approach, combining a LightGBM ML-model with a chemical clustering technique and subject-matter expertise to develop a model that use (up to) 45 proxy parameters to estimate orthophosphate at locations where other water quality parameters are already being measured.

- **Data inputs:** Our final model used 24 years of historical data collected by the Environment Agency (Water Quality Archive). This includes more than 3,000 different determinands, collected at more than 60,000 sampling sites. To ensure that only relevant samples are used, we decided to use only samples from rivers and surface water. Furthermore, following several stakeholder discussions, a decision was made to cap the input data to 0.5 mg/L, to exclude outliers. However, this decision might have influenced the model's ability to estimate orthophosphate in cases with high levels of orthophosphate. Therefore, we recommend that this cap is revised in further work.

Pilot catchments:

- **National dataset:** Orthophosphate data are available in low frequency (typically, 1 to 2 samples per month for most sites) for England (EA Water Quality Archive). This has been used to train our model across all EA sampling points in England. We were not able to get access to any high-frequency datasets during the project.

Outputs:

- **Output format**: Using up to 45 determinands (see model output report for details), the model estimates the level of orthophosphate as P in mg/l.

3.2 Theme 2: Optimising catchment monitoring

For stakeholders to decide what interventions are needed where to improve waterbody health, an understanding of the current catchment 'situation' must be known. Whilst many catchments have some level of existing water quality data, it can be challenging to understand the accuracy and reliability of existing water quality datasets within a catchment. Consequently, before a water quality intervention can be implemented, most catchment stakeholders must undertake preliminary water quality monitoring programmes, so that the decision about where to invest is informed by accurate and reliable data.

Designing and implementing preliminary water quality monitoring programmes is often time consuming and expensive. In many cases, multiple in-person site surveys are required during the initial stages of a project, to help inform where water quality sampling or monitoring 'schemes' should be implemented within a catchment. A single in-person site walk-over will generally take; 'about two days to do 500 metres of river' and can cost 'upwards of £200 per day' (C2, Interviews). As such, projects within larger catchment areas will incur significant financial costs before any actual water quality monitoring has begun.

Furthermore, continuous water quality monitoring is costly, limiting the number of monitors that can be deployed across a catchment. Given that water quality variables can fluctuate significantly over short timescales and across different spatial locations, monitoring schemes often lack the precision needed to capture these changes accurately. While continuous monitoring captures changes over time, it often misses spatial variability across different river sections, especially those distant from monitoring stations. Drones, conversely, can survey spatial changes effectively, but lack the temporal continuity needed for tracking water quality dynamics over time. To generate accurate and comprehensive water quality insights, stakeholders are depending on cost-effective, optimised monitoring schemes that capture the complex dynamics of entire catchments – addressing challenges of limited spatial and temporal coverage.

This theme has two core aims:

1.) To use remote sensing and AI/ML techniques to identify pollution sources that we do not typically have a lot of information about, or pollution sources that are typically more challenging to pinpoint without catchment walkovers.
2.) To develop a high-level, transparent water quality risk mapping framework, to help stakeholders understand where water quality risk is highest within a catchment and what the likely sources of water quality pollution are at any point along the river system.

Stakeholders

- **Water Companies:** to enable quick, reliable and efficient understanding of water quality risk within a catchment, to assist in the implementation of water quality interventions. Relevant for catchment-based approaches and catchment nutrient balancing efforts across both the wastewater and clean water sectors.
- **NGOs:** help build a clear understanding of water quality risk within catchments, to help target areas that could benefit from citizen science projects or additional water quality monitoring schemes.
- **Regulators:** help build a clear understanding of water quality risk within catchments, to help identify whether additional monitoring schemes may be needed to meet regulatory requirements.

Challenges

The primary challenges this theme aims to address is the significant financial costs associated with the design and implementation of a water quality monitoring schemes by stakeholders. These financial costs can be loosely separated into two themes:

(1) The cost of preliminary site surveys and the design of the monitoring scheme.
(2) The capital and operational cost of deployment and maintenance of continuous monitoring equipment or cost of undertaking regular sampling activities. According to Xylem, the typical cost for the purchase and installation of a permanent water quality monitor is around £20,000 per site, along with a cost of about £1000 per month for maintenance.

As the capital and operational costs associated with *in-situ* water quality monitors can be high, most catchment monitoring schemes rely on a relatively small number of monitoring sites. Consequently, stakeholders can also face challenges when attempting to understand water quality 'in between' monitoring sites. In other words, the interpolation of water quality data is often inaccurate and unreliable, making it difficult to understand how water quality varies within a waterbody, unless a significant number of monitors are installed. Furthermore, if monitors are in the 'wrong' location, the usability and quality of data they produce can be severely limited, minimising their cost-effectiveness.

Opportunities

The main aim of this theme is to use AI/ML and remote sensing to help reduce the financial costs associated with designing and implementing a water quality monitoring scheme within a catchment. The outputs from this theme should help

relevant stakeholders understand where water quality risk is highest within a catchment.

This has led to the definition of two building blocks, that could be used to accomplish the overall aim:

- **Building block 2A:** Use AI/ML and remote sensing to detect potential sources of pollution.
- **Building block 2C:** To develop a high-level water quality risk mapping framework to help inform catchment monitoring and intervention plans.

Building Block 2A: Use AI/ML and remote sensing to detect potential sources of pollution.

Defined problem and value:
Understanding potential sources of pollution in a catchment currently requires time-consuming on-the-ground walkovers or extensive work with GIS data. The aim of this building block is to use AI/ML and remote sensing to identify and map pollution sources in a time efficient way. Following several workshops and discussions with TAG members and project partners, three core model concepts were decided upon:

i) To use AI/ML to detect circular agricultural slurry tanks.
ii) To use AI/ML and remote sensing to detect bare cropland at sub-annual timescales.
iii) Use AI/ML and remote sensing to detect poached land.

These three potential sources of pollution were chosen, as they are typically more challenging to obtain data about, without the use of time-consuming, expensive, in-person catchment walkovers. Although other types of slurry storage are present across the UK, circular slurry tanks were decided upon by discussions with stakeholders, as they are likely the most common slurry storage structure type and have a clearly defined shape/structure that will aid AI/ML object detection techniques.

Technical specifications:

- **Modelling techniques:** Remote sensing and computer vision (AI/ML) that automates the process of finding potential sources of pollution.
  - Slurry tanks: YOLOv8 with an AdamW optimiser considering seven additional classes.

- Bare cropland: Bare Soil Index (BSI), Soil-Adjusted Vegetation Index (SAVI), Normalized Difference Moisture Index (NDMI), and Normalized Difference Snow Index (NDSI).
- Poached land: AI upscaled satellite imagery and a random forest classifier, which has been trained on manually digitised polygons across five land cover classes (poached land, bare cropland, cropland, trees and waterbodies).

- **Data inputs:** satellite imagery, land cover datasets, water company asset data, agricultural asset data, crop cover maps and BGS datasets.
    - Limitations of remote monitoring data (e.g. cloud cover, frequency of data, etc.) are considered.
    - Manual labelling of imaging used during some validation processes.

Pilot Catchments:

- All three models developed as part of building block 2A were trained on non-catchment specific datasets and can be applied anywhere the user has data inputs for (i.e. primarily satellite imagery).
    - Slurry tanks: Validated using 233 high-resolution satellite images across England, Wales and Denmark (due to readily available labelled slurry tank dataset).
    - Bare cropland: Validated in 19 catchments across England, Wales and Scotland.
    - Poached land: Tested on several images taken from UK catchments.

Outputs:

- For each of the three models, the primary output is a GIS layer containing geo-referenced pollution sources. The bare cropland model can also produce timeseries data about changes in bare cropland over time, which can be exported as a csv file.
    - The building block could be used to help develop a standard approach to desktop surveys.

Use cases:

The real-world application for stakeholders is for the models within this building block to enable:

- **Regulators:** Having a model to cheaply and efficiently identify potential sources of pollution in a catchment will help regulators to gauge what the key risks to waterbody health are within a catchment, without the need for extensive in-person surveys. In addition, the model outputs may help identify pollution sources that have not previously been accounted for. Utilising this model as a standard approach for desktop surveys would allow for more

comparability between different catchments, helping to highlight key areas of risk.

- **Water companies and charities**: Having a model that identifies potential pollution risks within a catchment will enable water companies and charities to save time and money associated with extensive in-person catchment surveys. The model outputs will give stakeholders an understanding of potential 'high-risk' zones and highlight what issues need to be addressed with water quality interventions.

Building Block 2C: To develop a high-level water quality risk mapping framework to help inform catchment monitoring and intervention plans.

Defined problem and value:
Stakeholders can face significant time and financial outgoings when attempting to understand the location and severity of water quality issues within a catchment. To reduce these costs, this building block aims to produce a high-level, transparent risk mapping framework, that allows users to understand where water quality risk is highest within a particular catchment. The risk map focusses on two water quality determinands: ammonia and phosphate. These two pollutants were chosen by the project team and wider stakeholders for two primary reasons: 1.) understanding catchment nutrient loading is of significant interest to most stakeholders (including WASCs, eNGOs, regulators, etc.), and 2.) the source of these pollutants is varied, allowing the model development to include several different pollutant sources within the model (e.g. WWTW effluent, CSOs, agricultural runoff, urban runoff, etc.).

Technical specifications:

- **Modelling techniques:** The model is built up using individual 'risk layers', each of which represents a specific source of water quality pollution. Water quality risk is calculated using open-datasets and high-level assumptions. Water quality risk methodology is transparent, allowing users to justify any decisions or investments made using the tool. Water quality 'risk-degradation' / 'risk-transport' mechanisms are incorporated into the map, to allow users to understand how the risk from different pollution sources changes as you move through the catchment.

- **Data inputs:** Discharge permits, CSO event duration monitoring (EDM), livestock and agricultural datasets, DEM, land cover data, river morphology, terrain, EA WIMS data, etc.

Pilot Catchments:

- The risk mapping tool can be automatically run for any catchment within England.

Outputs:

The output produced from this building block is a clear visual representation of annualised water quality risk within a catchment. The model is quick and easy for users to run, with minimal need for pre-processing of data. The risk map should provide users with an understanding of source apportionment throughout the catchment. The risk map outputs are easily exportable into GIS platforms.

Use cases:

The real-world application for stakeholders is for the models within this building block to enable:

- **Regulators, water companies and NGOs:** to have a clear, high-level understanding of annual-average water quality risk within a catchment, to help ensure that regulators, water companies and charities understand where water quality interventions or monitoring is most needed. The open, transparent nature of the model should allow for appropriate justification of investment decisions, enabling regulators, water companies and charities to work more effectively together.

Following on from this project, outputs from 2C could be developed further. Possible use cases include:

- The addition of more water quality determinand layers.
- The addition of more individual risk layers (i.e. sources of pollution).
- Improved flow/dilution estimations, to improve water quality risk estimations – possibly bringing in external monitored or simulated flow, like from the RDMAI Open Flow Model.

"The River Deep Mountain AI (RDMAI) project has provided the important opportunity to examine some of the potential applications of Artificial Intelligence (AI) and advanced data analytics, to the datasets we have available in the water sector. We have advanced the capability of partners to design credible models and approaches; it is crucial to continue this work to further develop Explainable AI approaches for application across the sector."

Michelle Walker, Technical Director, The Rivers Trust.

3.3 Theme 3: Getting more value out of continuous monitoring

To better understand the impact of discharges from sewage treatment works and storm overflows on waterbodies, the Department of Environment, Food and Rural Affairs (Defra) has put a statutory requirement on English water companies to continuously monitor upstream and downstream of their assets (Environment Act 2021, Section 82).

This will increase current OPEX/CAPEX (capital and operating costs) related to monitoring and create unprecedented levels of incoming real-time data for both water companies and regulators.

This theme is initially most relevant for water companies and regulators in England, but we assume that moving forward, more stakeholders will increase their capabilities for continuous monitoring of water quality, making it important for all to sense-check vast amounts of data.

Stakeholders

- **Water companies**: Defra's provisional guidance on implementing section 82 of the Environment Act (2021) requires water companies in England to publish continuous monitoring data within 1 hour. This puts additional pressure on water companies to sense-check data.
- **Regulators**: Like water companies, regulators are expected to publish and use continuous water quality data coming in from water companies.

Challenges

- Increased capital cost of acquiring and deploying sensors (as much as £20,000 per site for continuous water quality monitoring station), as well as setting up live dashboards, visualisations etc. to adhere to statutory requirements.
- Increased operational cost to maintain sensors (can be upwards of £1000 per month for sensor maintenance).
- Unprecedented amounts of incoming data that must be sense-checked and published within 1 hour every day, to align with Section 82 of the Environment Act 2021. Based on unofficial calculations, over 1.25 billion data points may be generated annually, assuming each sonde collects data for 5 parameters at an hourly timestep. If this timestep is reduced to 15 minutes, this number will be increased to near 5 billion data points.
- Together, these put water companies and regulators under pressure to handle sensors, data and sense-checking.

Opportunities

Our main hypothesis for this theme is that by utilising AI and ML, we can support water companies in adhering to the provisional guidance on implementing section

82 of the Environment Act (Defra, 2023) where water companies are required to publish continuous water quality data within 1 hour.

'Let's just talk about the data for a minute. We have to publish these results within one hour. On a customer facing dashboard [...] To do that, we need to be confident that the results are accurate, that the results, there's no false flags in there. We need to screen it.' (Water Company Interviews from Phase 1)

[...] 'We can do it with humans or current technology, but definitely the world of AI will play a massive part in achieving that one-hour publication of that data with confidence.' (*Water Company Interviews from Phase 1*)

We have defined one specific opportunity where AI/ML can be applied to overcome some of the specific challenges related to this theme.

- **Building block 3A:** Detecting and clustering anomalies in continuous monitoring data.

Building Block 3A: Detecting and clustering anomalies in continuous water quality monitoring data.

Defined problem and value:

- Water companies and regulators are expected to publish continuous monitoring water quality data every hour adhering to the provisional guidance set out by Defra on how to implement section 82 of the Environment Act (Defra, 2023).
- *'Water quality data must be made publicly available in near real time (within one hour), and in a common format across England.'*[2]
- This puts additional pressure on water companies, and regulators, to sense-check data quickly before publishing it to the public.
- This building block and theme is focussed on the ability to rapidly assess water quality timeseries data.
- An AI/ML model that can detect and flag anomalies, based on a baseline understanding of a given catchment. It has the potential to be used to help speed up the process of analysing section 82 data and reduce the risk of publishing erroneous and misleading data.
- Following stakeholder discussions, in this context, anomalies refer to any outliers in the data that do not fit within the normal fluctuations and trends at a particular location. This could include pollution events, sensor malfunctions,

[2] Defra (2023). Continuous Water Quality Monitoring Programme – Provisional technical guidance for sewerage undertakers on implementing s.82 of the Environment Act 2021

or other things that might produce an outlier in the data. The purpose of this building block is not to categorise anomalies, but to simply detect them.

Technical specifications:

- **Modelling techniques:** AI/ML model that makes use of existing water quality data to detect and flag anomalies and outliers in incoming continuous monitoring data.
- A significant challenge associated with theme 3 was the removal of seasonal variation within the water quality. Significant input and discussion from project-partners and stakeholders helped determine the best method(s) for doing this, which can be found in the model documentation on Github.

- **Data inputs**
  - Section 82 parameters: ammonia, dissolved oxygen, turbidity, pH and temperature. The model has been designed to analyse relatively high frequency data (e.g. hourly). Data should be saved in the specified csv format, as shown in the relevant model documentation.

Pilot catchments:

  - The model has been developed so that it can be applied to any data timeseries, provided the correct water quality parameters are included. The model is not limited to a specific catchment. Model validation is undertaken using existing water quality timeseries data from the EA and manually labelled water quality timeseries.

Outputs:

Detect and flag anomalies in the data that requires human verification. Anomalies will then be clustered, based off their characteristics.

Use cases:

The real-world application for stakeholders is for the models within this building block to enable:

- **Regulators:** Automated outlier detection enables **regulators** to understand where and when river water quality parameters are out of the norm. This could help highlight potential pollution events and prompt any further investigation. The targeted approach improves resource allocation, improved regulatory enforcement, and increased quality and reliability of the data shared with the public. The sheer volume of section 82 type data that will be collected over the next AMP cycle means automatic detection of anomalies will become critical.
- **Water companies:** Automatic anomaly detection of pollution events of system malfunctions enables **water companies** to conduct rapid analysis of

times series data, to help understand the environment and/or water quality monitoring infrastructure. Being able to flag issues quickly and systematically will improve data quality, streamline compliance and demonstrate transparency, aligning with the Environment Act´s standards

- **eNGOs:** Environmental charities access to trustworthy near real-time information about the health of their waterbodies and prioritise conservation efforts and focus resources effectively. This capability strengthens the impact of charities by enhancing their data-driven advocacy, resource allocation and engagement with water quality initiatives.

"River Deep Mountain AI enables a more open way of working, opening up complex river data so insight is shared, transparent, and shaped collaboratively. It lowers barriers between expertise, enabling collective learning and making evidence led collaboration the default rather than the exception"

Lauren Isbister,
Mainstreaming Nature Based Solutions Project Manager, South West Water

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4 Summary and Conclusion

Throughout the RDMAI project, a continual effort was made to ensure that stakeholders from across the water sector had the opportunity to input into model development, so that the outputs from each of our models remained relevant to all interested parties. Wider stakeholder sharing sessions were undertaken quarterly, to present an overview of progress, provide a platform for stakeholder feedback and ensure that machine learning objectives were validated with relevant stakeholders, aligning with deliverables T2.5 and T2.6 that were documented at the beginning of this report.

Appendix A

A breakdown of attendance by sector for each of the demo days and the showcase event can be found below.

16th December 2024:

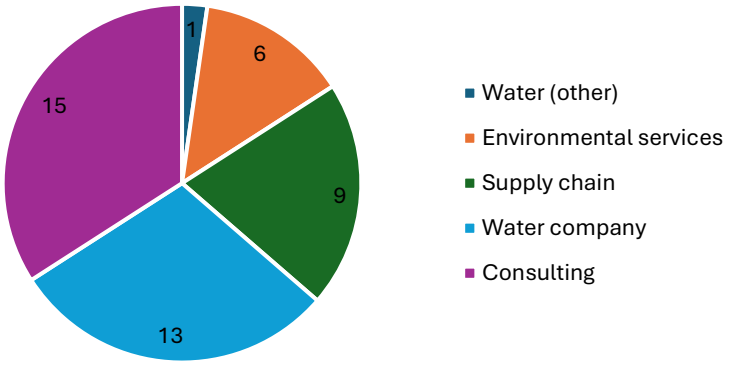

Figure 2 – Attendance at the first demo day, summarised by sector.

6th March 2025:

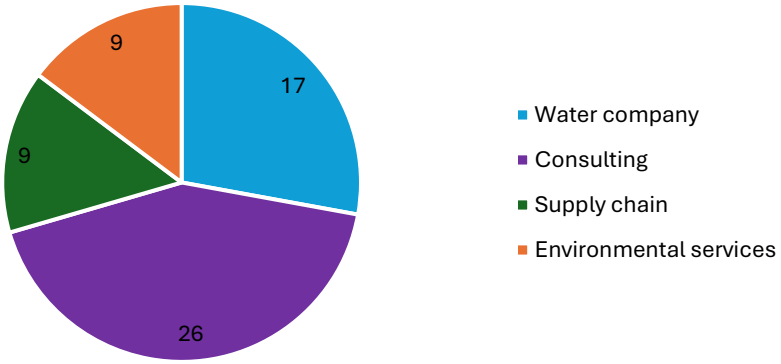

Figure 3 – Attendance at the second demo day, summarised by sector.

30th April 2025:

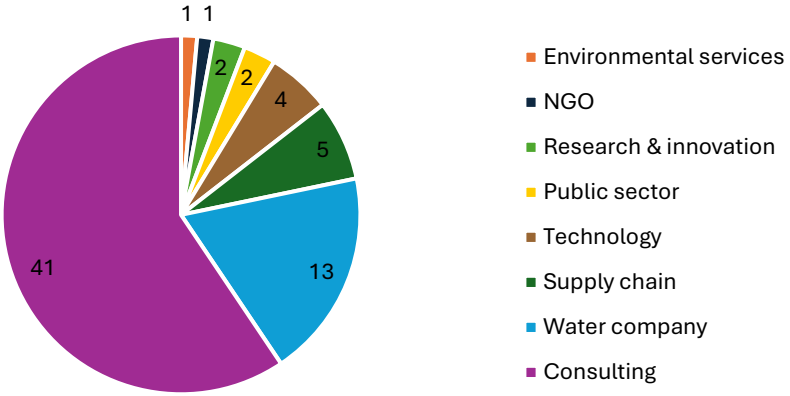

Figure 4 – Attendance at the third demo day, summarised by sector.

25th November 2025:

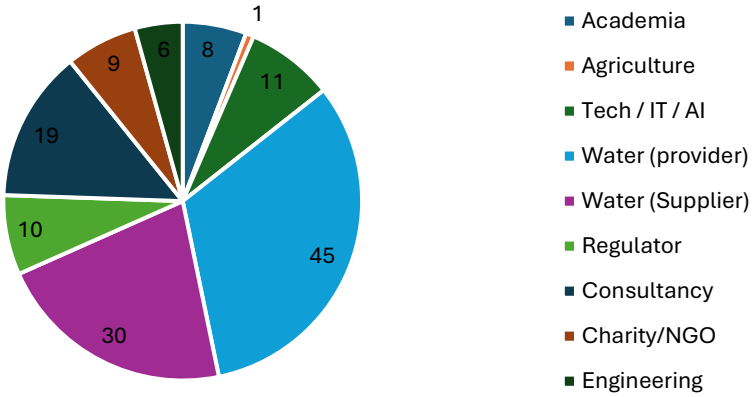

Figure 5 – Attendance at the showcase event, summarised by sector.

Appendix B

A summary of data availability is displayed for each of the originally suggested pilot catchments during phase 1 of the RDMAI project.

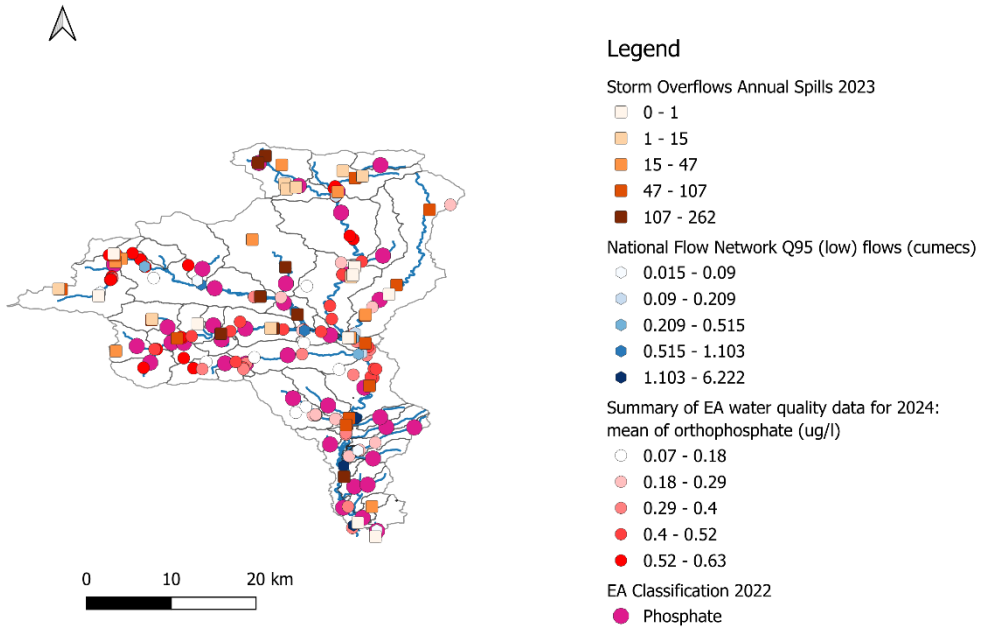

Figure 6 – Avon Hampshire

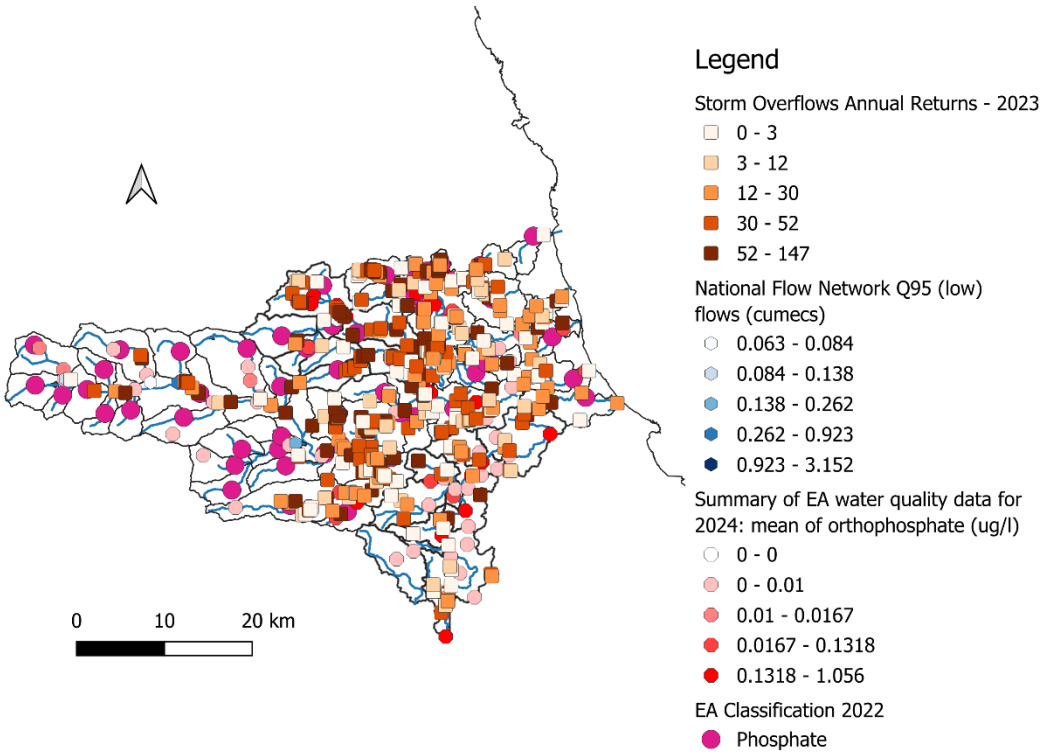

Figure 7 – Browney

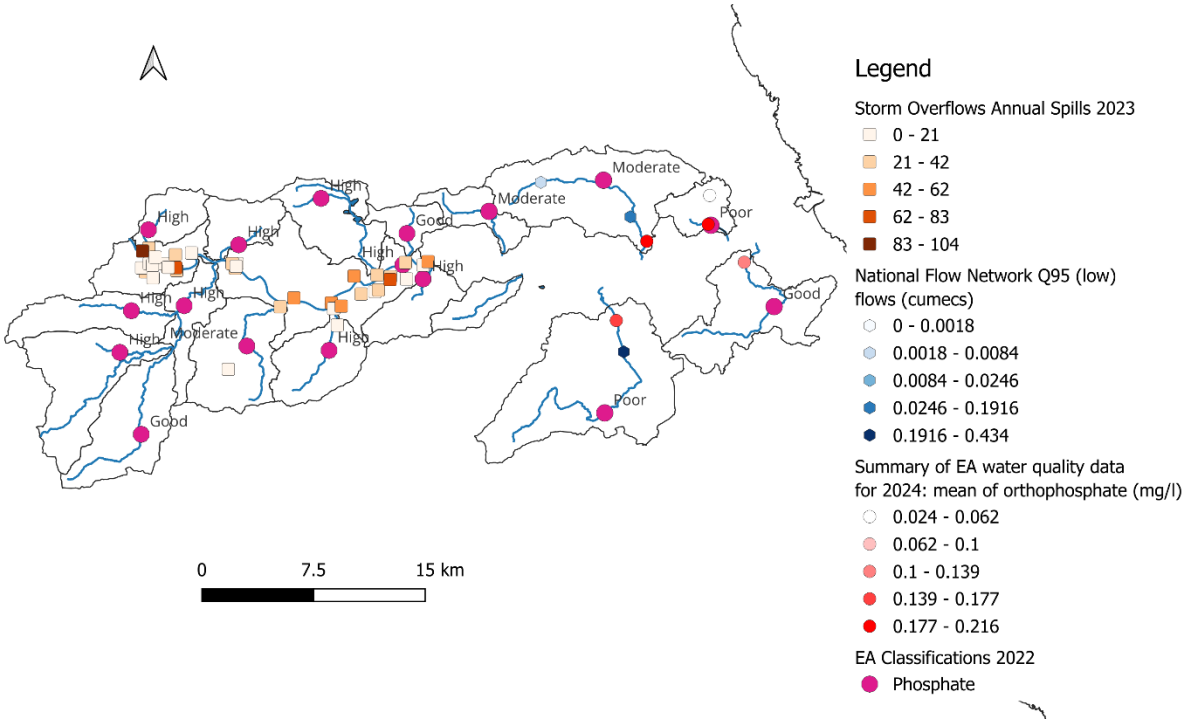

Figure 8 – Cullercoats

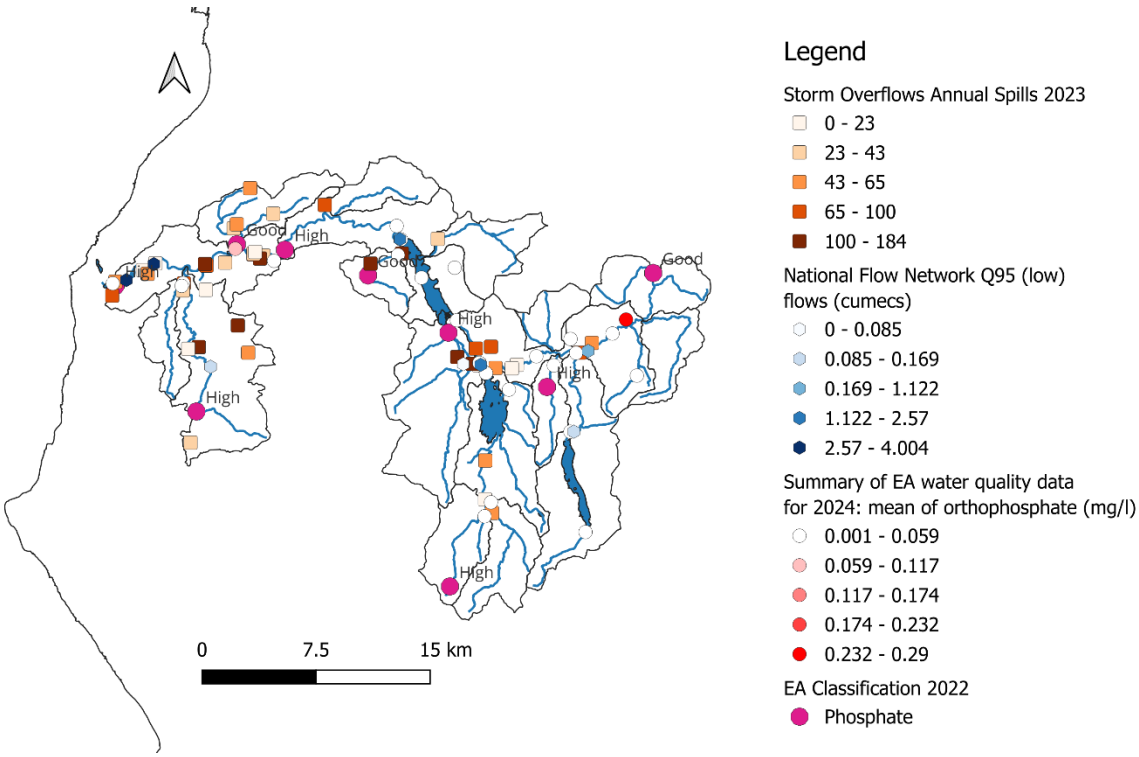

Figure 9 – Derwent

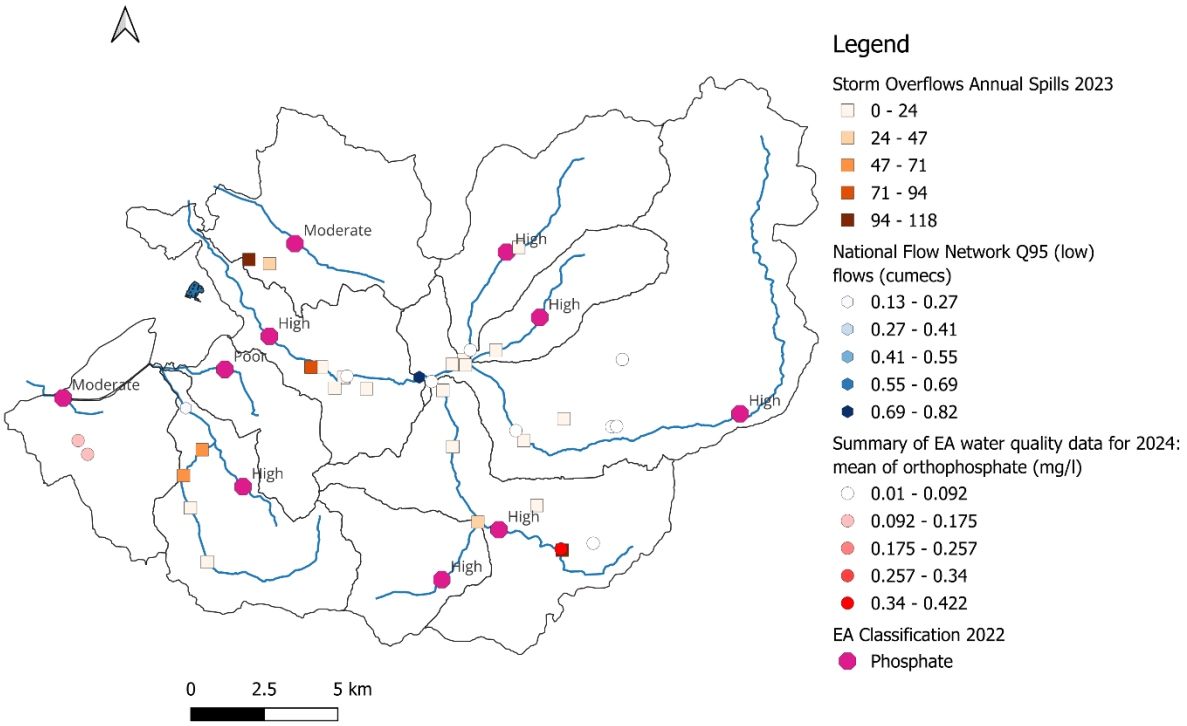

Figure 10 – Frome and Cam

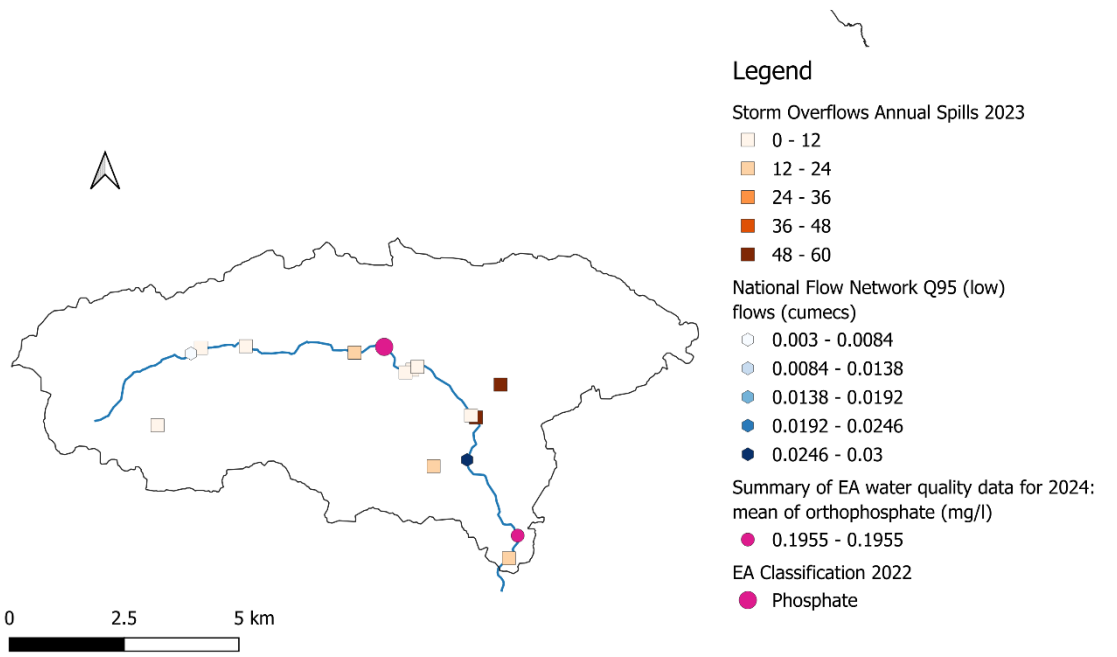

Figure 11 – Ouseburn

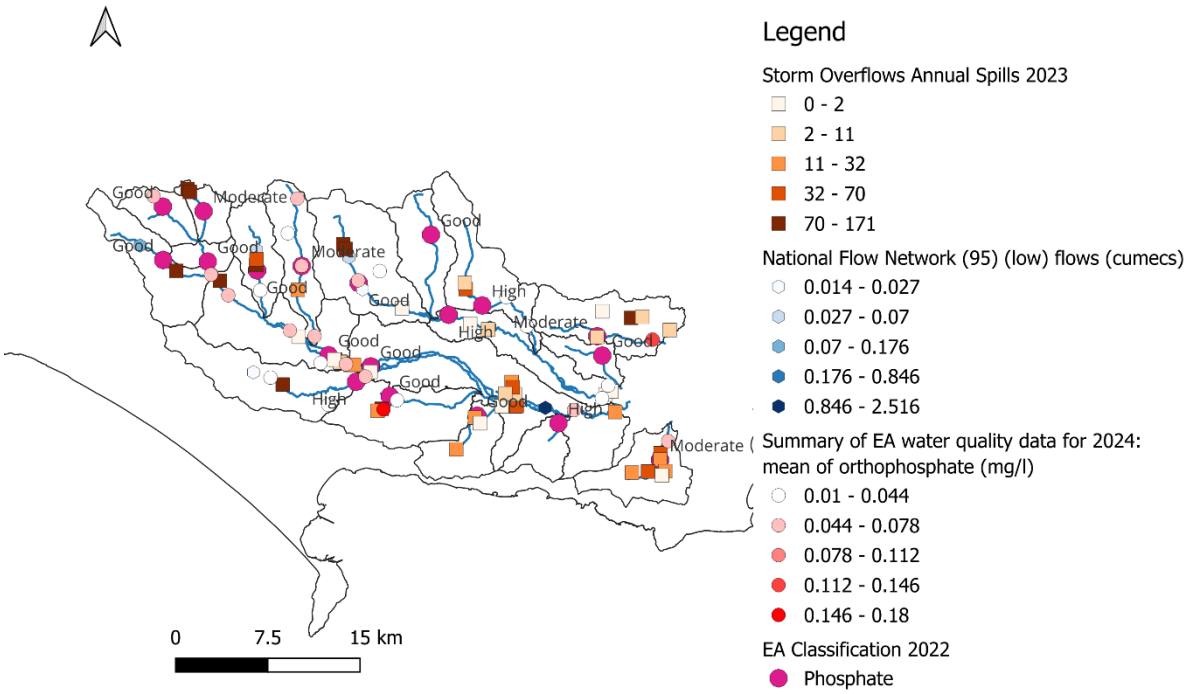

Figure 11 – Poole Harbour

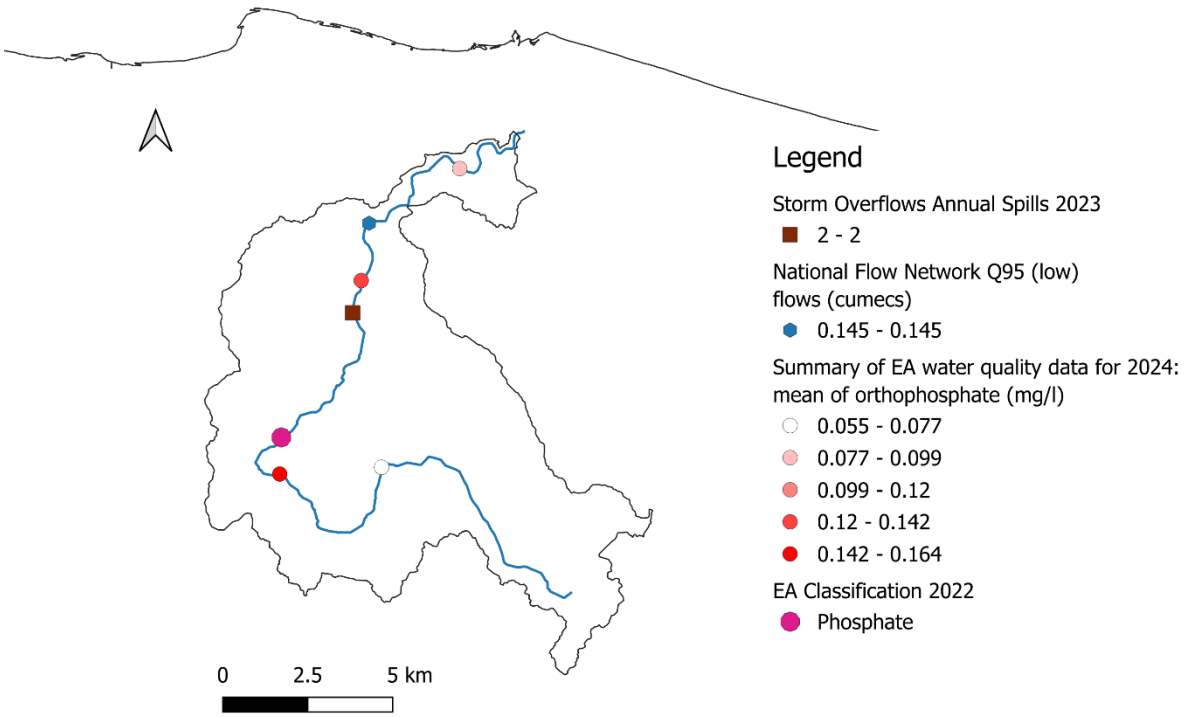

Figure 11 – Stiffkey

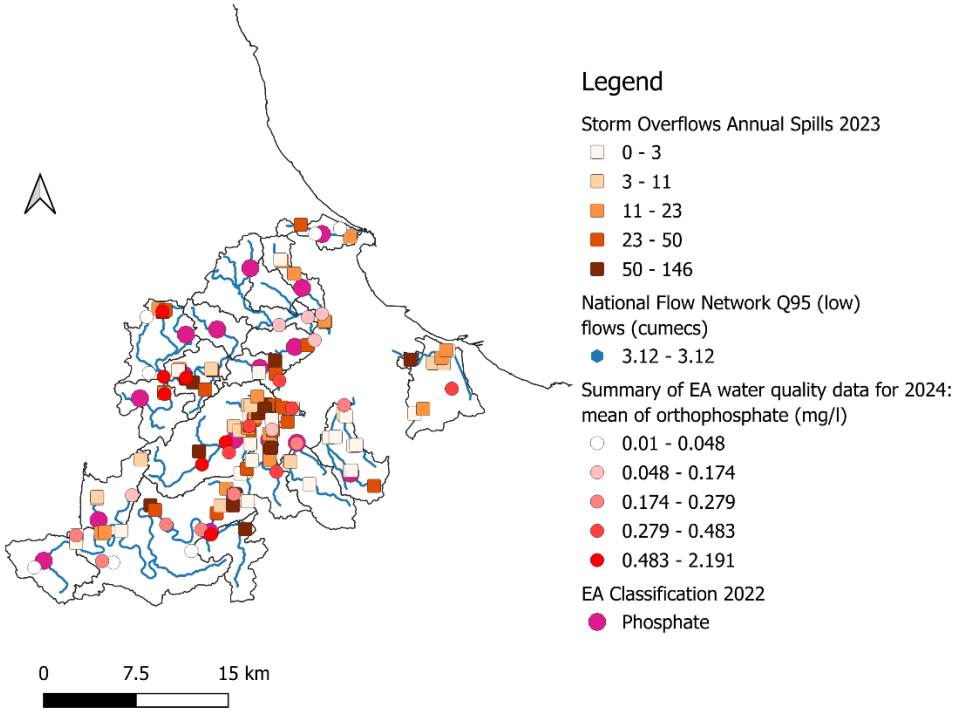

Figure 11 – Tees

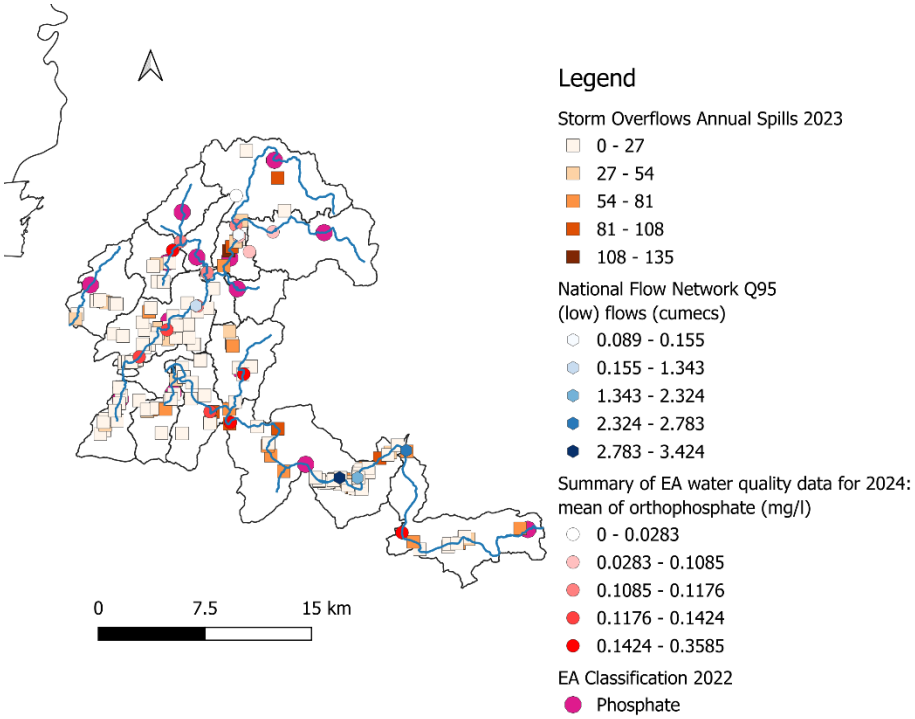

Figure 12 – Warleigh Weir

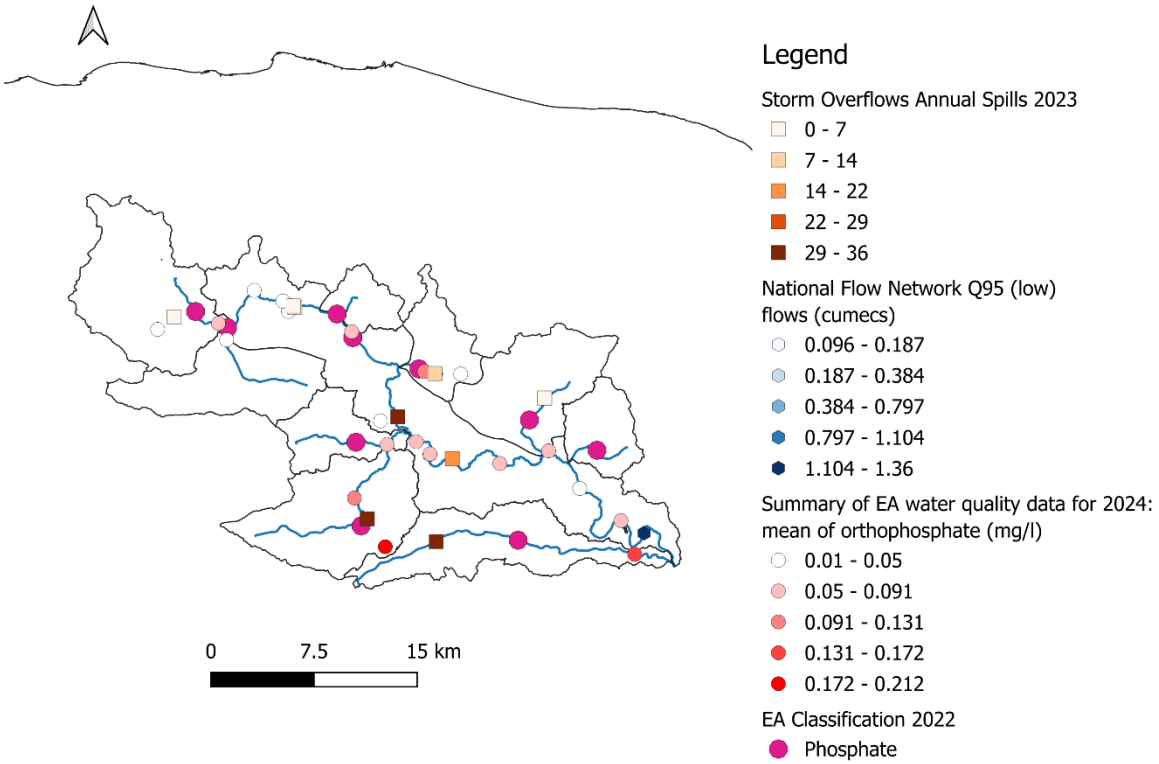

Figure 11 – Wendling Beck